

Image Analysis Report

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Executive Summary

This piping assembly exhibits severe atmospheric corrosion and widespread coating system failure across the central cross and flange connections. The advanced state of material degradation on fasteners and the presence of thick, stratified rust scale suggest a high risk to both structural and pressure-containing integrity.

Key Findings

#	Finding
1	Advanced stratified rust scale on the upper flange assembly indicating significant material loss.
2	Widespread coating delamination and peeling on the horizontal pipe runs and central body.
3	Heavy corrosion on fasteners with visible reduction in bolt head geometry on the left and right flanges.
4	Localized pitting and general surface metal loss on the central cross-fitting where the coating has failed.
5	Potential crevice corrosion at the flange-to-pipe and flange-to-flange interfaces.
6	Internal corrosion state and remaining wall thickness cannot be determined from this visual inspection alone.

Localized detections (VLM, approximate)

- Severe Flange Scale (conf. 95%): x=0.330, y=0.050, w=0.350, h=0.220
- Fastener Degradation (conf. 92%): x=0.160, y=0.230, w=0.160, h=0.630
- Fastener Degradation (conf. 92%): x=0.680, y=0.220, w=0.170, h=0.630
- Coating Failure (conf. 88%): x=0.840, y=0.320, w=0.160, h=0.400
- General Metal Loss (conf. 90%): x=0.280, y=0.250, w=0.420, h=0.600

Recommendations

#	Recommendation
1	Perform Ultrasonic Testing (UT) to verify remaining wall thickness of the pipe and fitting against the design minimum.
2	Mechanically clean all surfaces to SSPC-SP 10 (Near-White Blast Cleaning) to reveal the true extent of pitting.
3	Replace all corroded bolts and nuts with new, high-grade coated fasteners (e.g., fluoropolymer or zinc-plated).
4	Apply a high-performance multi-coat industrial protection system, including a zinc-rich primer and epoxy mid-coat.
5	Inspect flange face sealing surfaces for corrosion damage once the joints are disassembled for bolt replacement.
6	Increase the frequency of visual inspections to quarterly until the remediation project is completed.